

Answer Key — How Life Is Sustained on Earth

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **oxygen** — photosynthesis releases oxygen.
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- Q2** (b) **respiration** — respiration uses O_2 and releases CO_2 .
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- Q3** (c) **70%** — about 70% of Earth's surface is water.
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- Q4** (b) **keeping Earth warm enough for liquid water** — it keeps temperatures right for liquid water.
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- Q5** (a) **Both true, R explains A** — plants release the oxygen animals breathe — as R explains.
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- Q6** **photosynthesis** — this food-making process is photosynthesis.
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- Q7** **respiration** — respiration uses oxygen, releases CO_2 .
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Short answer

- Q8** Photosynthesis takes in CO_2 and releases O_2 ; respiration uses O_2 and releases CO_2 . Together they recycle the two gases, keeping the air suitable for life.
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- Q9** Water is essential for all life processes; it covers about 70% of Earth's surface (making Earth the Blue Planet).
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- Q10** It supplies oxygen for respiration, and its mild greenhouse effect keeps Earth warm enough for liquid water (and the ozone layer blocks UV).
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Long / application

- Q11** Sunlight lets plants carry out photosynthesis using water (from soil) and carbon dioxide (from air) to make food, releasing oxygen. Animals and plants use this oxygen for respiration, releasing CO_2 that plants reuse. Sunlight also warms Earth, and the mild greenhouse effect keeps water liquid. Thus air, water and sunlight are linked in cycles that keep life going.
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- Q12** Habitability factors: the right distance from the Sun (liquid water/habitable zone), the right size (gravity to hold an atmosphere), an atmosphere with oxygen and an ozone layer, and a magnetic field that shields from harmful particles. Sustaining cycles: photosynthesis and respiration recycle oxygen and carbon dioxide, water cycles through the environment, and sunlight powers life — all interconnected, which is why we must protect Earth's balance.
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